

Assignment 4: Tool Specification

Personal Productivity AI Agent

AI Engineering Fellowship 2026 – Week 3

Student: Hassan Murtaza Zaidi

Tool Specification Table `create_task`

Purpose: Create a new task

Input: title (required), description, priority, due_date, tags

Output: Task object

Operation: Write

Approval Required: Only bulk create

Possible Errors: Invalid fields

Example Call: `create_task(...)`

Example Result: Success response with structured JSON.

`list_tasks`

Purpose: List tasks

Input: status, priority, tag

Output: Task list

Operation: Read

Approval Required: No

Possible Errors: None

Example Call: `list_tasks(...)`

Example Result: Success response with structured JSON.

`update_task`

Purpose: Update task

Input: task_id(required), updates(required)

Output: Updated task

Operation: Write

Approval Required: Yes

Possible Errors: Task not found

Example Call: `update_task(...)`

Example Result: Success response with structured JSON.

`complete_task`

Purpose: Complete task

Input: task_id(required)

Output: Completed task

Operation: Write

Approval Required: Yes

Possible Errors: Task not found

Example Call: `complete_task(...)`

Example Result: Success response with structured JSON.

delete_task

Purpose: Delete task

Input: task_id(required)

Output: Confirmation

Operation: Write

Approval Required: Yes

Possible Errors: Task not found

Example Call: delete_task(...)

Example Result: Success response with structured JSON.

save_note

Purpose: Save note

Input: title(required),content(required),category,tags

Output: Saved note

Operation: Write

Approval Required: No

Possible Errors: Missing content

Example Call: save_note(...)

Example Result: Success response with structured JSON.

search_notes

Purpose: Semantic search

Input: query(required)

Output: Matching notes

Operation: Read

Approval Required: No

Possible Errors: No results

Example Call: search_notes(...)

Example Result: Success response with structured JSON.

extract_meeting_actions

Purpose: Extract actions

Input: meeting_notes(required)

Output: Summary/actions

Operation: Read

Approval Required: No

Possible Errors: Empty notes

Example Call: extract_meeting_actions(...)

Example Result: Success response with structured JSON.

generate_work_plan

Purpose: Generate work plan

Input: available_hours,date,user_priorities

Output: Daily plan

Operation: Read

Approval Required: No

Possible Errors: Invalid input

Example Call: generate_work_plan(...)

Example Result: Success response with structured JSON.